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Cluster of Excellence PhenoRob, University of Bonn

Motivation letter: Junior Research Group Leader (application code 2026/397)

Dear Professor Kuhlmann and members of the selection committee,

I am applying for one of the Junior Research Group Leader positions in PhenoRob. I am finishing my PhD in Mechatronics and Robotics at Aalborg University (defence expected in 2027), after an MTech in Mechatronics and a BTech in Mechanical Engineering. My work is on robots that physically touch living things, and I would like to build a group that brings that know-how to the field: robots that handle plants gently and measure them through contact.

In my PhD I designed a hybrid-actuated shoulder exoskeleton, where a spring and a small 6 Nm motor share the load. It cut energy use by more than 25% and won the Best Research Paper Award at IFToMM ISRM 2026. On a 5-bar parallel robot I used variable-stiffness actuators and the robot Jacobian to estimate contact force without a sensor at the tip, within 10% RMS error of a force sensor, and we used it to measure human arm impedance in the gait lab. During my Master's at CSIR-CMERI I fabricated pneumatic artificial muscles and worked on the PID loop of a gain-scheduled controller that brought tracking error down to 0.3-0.78% (IEEE CONECCT 2022). I build my own prototypes and test rigs, and I have run experiments with 12 human participants.

The idea behind my group would be compliant, low-energy actuation and contact sensing for crop robots. Two PhD projects could start it. The first would estimate stem and leaf stiffness from the robot's own actuators while it touches the plant, giving a new mechanical trait for phenotyping without extra sensors. The second would design energy-efficient compliant end effectors for selective handling and harvesting, tested in PhenoRob's field trials. Both fit the Cluster's work on robotic sensing and new cropping systems, and I give more detail in the attached concept.

I will be honest that agriculture is a new field for me and that my publication record is still at an early stage. That is exactly why PhenoRob appeals to me: plant scientists, geodesists and roboticists sit in one cluster, and I would learn fast from them while bringing hardware and actuation skills that complement the existing perception and mapping strengths. I have supervised Bachelor's students since 2023, and I also draw. I hold a Diploma in Fine Arts and make my own figures and demonstrator visuals, which helps when explaining research to farmers, students and the public.

Thank you for considering my application. I would be glad to discuss the group concept with you.

Kind regards,
Arunabha Majumder